Real Analysis: Answer to Exercise 3

Due date:

Question 1. **Solution**: $|x-a|$ is the distance between x and a. If this distance is less than every positive number ϵ (or less than every $1/n$, where $n > 0$), then the distance must be 0. But if the distance from x to a is 0, then $x = a$.

Question 2. **Solution**: (a) This says exactly that $f(x) \to \ell$ as $x \to a$.

(b) This tells us that f is bounded on every interval $[-N, N]$; that is, for every $N > 0$ there is a $K > 0$ such that $-K \leq f(x) \leq K$ whenever $-N \leq x \leq N$. To see this, fix $N > 0$ and choose $\delta_N > 0$ so that $(a - \delta_N, a + \delta_N)$ contains $[-N, N]$. The condition holds for all $\delta > 0$, so for δ_N there is $\epsilon > 0$ with $|f(x) - \ell| < \epsilon$ for all $x \in (a - \delta_N, a + \delta_N)$, $x \neq a$; choosing K large enough to contain $f(a)$ and $(\ell - \epsilon, \ell + \epsilon)$ gives the claim.

(c) This tells us that f is bounded on all of $\mathbb{R}$; that is, there is a $K > 0$ such that $-K \leq f(x) \leq K$ for all x. Take the given $\epsilon > 0$ and choose K so that $[-K, K]$ contains $f(a)$ and $(\ell - \epsilon, \ell + \epsilon)$. For any $x \neq a$ there is $\delta > 0$ with $0 < |x - a| < \delta$, so $|f(x) - \ell| < \epsilon$, giving $-K < f(x) < K$.

(d) Let $\delta > 0$ be as given. Then whenever $x \neq a$ lies in $(a - \delta, a + \delta)$, the condition says that $|f(x) - \ell| < \epsilon$ for all $\epsilon > 0$. As in Question 1, this forces $f(x) = \ell$. So $f(x) = \ell$ for all x in $(a - \delta, a + \delta)$ except possibly at $x = a$.

Question 3. **Solution**: For the first limit, given $\epsilon > 0$, choose $\delta = \epsilon$. If $0 < |x - a| < \delta$ then $|x - a| < \epsilon$, as required. For the second, given $\epsilon > 0$, any $\delta > 0$ works, since $|c - c| = 0 < \epsilon$ for all x.

Question 4. **Solution**: (a) Given $\epsilon > 0$, choose $\delta = \epsilon/3$. If $0 < |x - 2| < \delta$ then $|(3x - 1) - 5| = |3x - 6| = 3|x - 2| < 3\delta = \epsilon$.

(b) Given $\epsilon > 0$, choose $\delta = \sqrt{\epsilon}$ (or $\delta = \min\{1, \epsilon\}$). If $0 < |x - 0| < \delta$ then $|x^2 - 0| = x^2 < \delta^2 = \epsilon$.

Question 5. **Solution**: Given $\epsilon > 0$, choose $\delta = \epsilon/2$. If $0 < |x - 3| < \delta$ then $|(2x + 1) - 7| = |2x - 6| = 2|x - 3| < 2\delta = \epsilon$.

Question 6. **Solution**: (a) Let $f(x) = \dfrac{3x^3 - 5x^2 + 7x - 13}{2x^3 - \pi}$. Dividing top and bottom by x^3,

$$f(x) = \frac{3 - 5/x + 7/x^2 - 13/x^3}{2 - \pi/x^3} \to \frac{3}{2} \quad \text{as } x \to \infty.$$

To prove this, let $\epsilon > 0$. Following the estimate in the lecture,

$$\left| f(x) - \frac{3}{2} \right| \le \frac{5}{|x|} + \frac{7}{|x^2|} + \frac{13}{|x^3|} + \frac{3\pi}{|2x^3|} \le \frac{50}{x}$$

for x large enough that $\pi \le x^3$. Let $X_\epsilon > \max\{50/\epsilon, \pi^{1/3}\}$; then $x > X_\epsilon$ gives $|f(x) - 3/2| \le 50/x < \epsilon$.

(b) Let $f(x) = 2x^2 - 3x + 5$. As $x \to 1$, $f(x) \to 2 \cdot 1^2 - 3 \cdot 1 + 5 = 4$. For $\epsilon > 0$,

$$|f(x) - 4| = |2x^2 - 3x + 1| = |(2x - 1)(x - 1)| = |2x - 1||x - 1|.$$

Taking $|x - 1| < 1/4$ gives $1/2 < 2x - 1 < 3/2$, so $|f(x) - 4| < \frac{3}{2}|x - 1|$. Choose $\delta < \min\{2\epsilon/3, 1/4\}$; then $|x - 1| < \delta$ gives $|f(x) - 4| < \epsilon$.

(c) Let $f(x) = 1/(1 - x)$. As $x \to 2$, $f(x) \to -1$. For $\epsilon > 0$,

$$|f(x) - (-1)| = \left| \frac{2 - x}{1 - x} \right| = \frac{|2 - x|}{|1 - x|}.$$

For $|2 - x| < 1/4$ we have $|1 - x| = |x - 1| > 3/4$, so $|f(x) + 1| < \frac{4}{3}|x - 2|$. Choose $\delta < \min\{3\epsilon/4, 1/4\}$; then $|x - 2| < \delta$ gives $|f(x) + 1| < \epsilon$.

(d) Let $f(x) = (\sin x)/x$. We claim $f(x) \to 0$ as $x \to \infty$. For $\epsilon > 0$,

$$|f(x) - 0| = \frac{|\sin x|}{x} \le \frac{1}{x}.$$

Let $X_\epsilon = 1/\epsilon$; then $x > X_\epsilon$ gives $|f(x)| \le 1/x < \epsilon$.

(e) Let $f(x) = x \sin(1/x)$. We claim $f(x) \to 0$ as $x \to 0$. For $\epsilon > 0$,

$$|f(x) - 0| = |x|\,|\sin(1/x)| \le |x|.$$

Choose $\delta = \epsilon$; then $0 < |x| < \delta$ gives $|f(x)| \le |x| < \epsilon$.